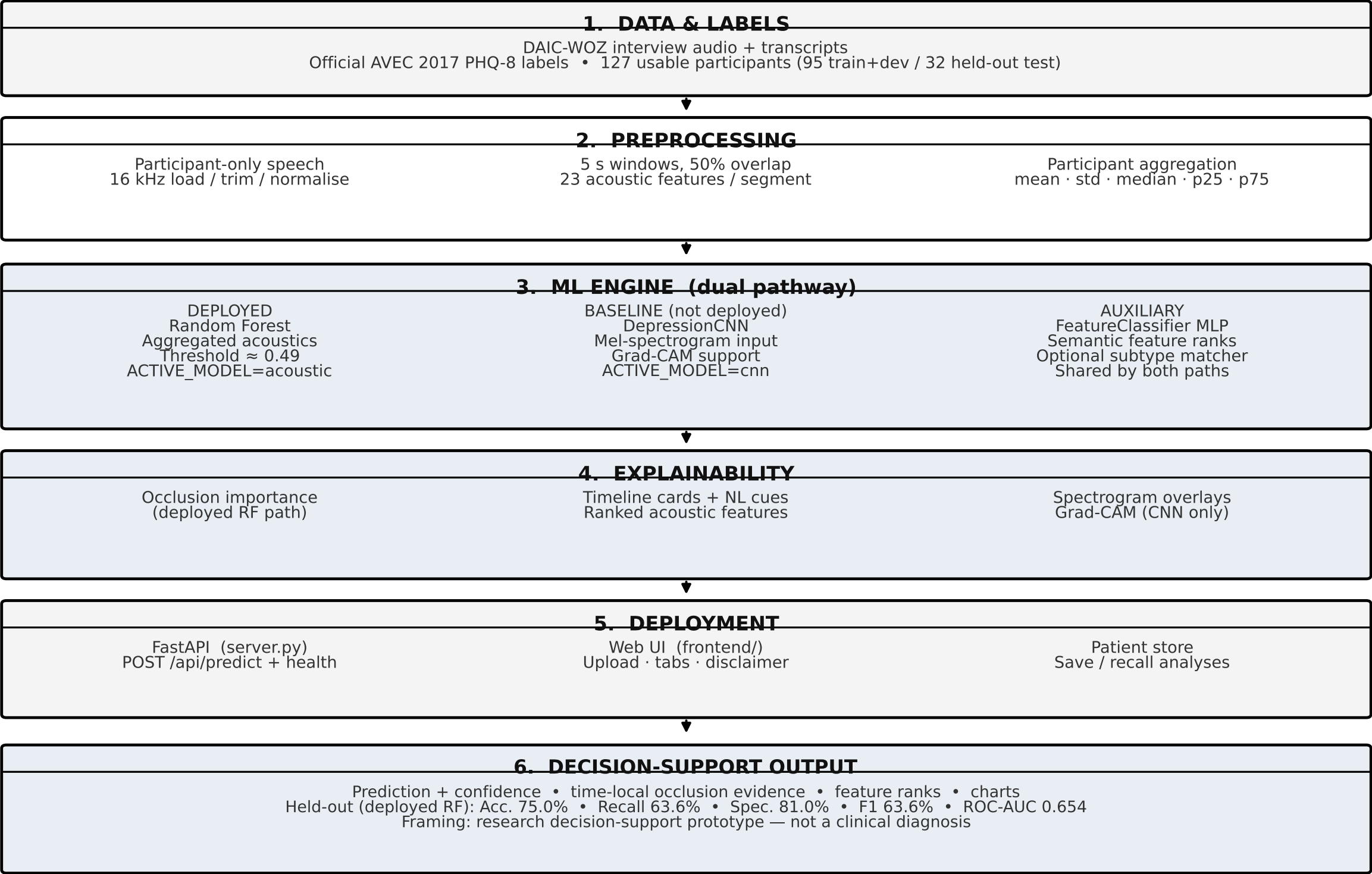


Figure 3.1 Architectural Summary Diagram

Explainable speech-based depression detection — dual pathway with deployed Random Forest



Previous Extra Trees candidate (not deployed): Acc. 59.4%, recall 63.6%. CNN + Grad-CAM retained as baseline only.

Deployed / key

Supporting

Leakage-aware protocol: official PHQ labels, speaker-independent split, threshold selected on train+dev only.